

Lightweight Multimodal Person Detection and Temporal Post-Processing for Aerial Search and Rescue

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Abstract—Autonomous unmanned aerial vehicles (UAVs) deployed in wilderness search-and-rescue (SAR) missions require reliable person detection under severe size, weight, power, and computing (SWaP-C) constraints. While multimodal sensing combining visible red-green-blue (RGB) and thermal infrared (TIR) cameras mitigates single-modality dropouts caused by canopy shade, darkness, or thermal crossover, frame-level object detectors frequently produce transient false positive spikes. In autonomous aerial operations, each false alarm triggers wasteful loiter maneuvers, battery depletion, and spurious transmissions over bandwidth-constrained Internet of Things (IoT) satellite or cellular links. This paper presents a systematic comparative evaluation of causal temporal post-processing methods designed to convert continuous multimodal confidence streams into reliable alarm decisions while suppressing false alarms. We deploy the lightweight YOLO11n detector (2.6 million parameters, 6.5 GFLOPs at 640×640 resolution) trained with batch size 16 in single-precision floating-point (FP32) arithmetic to guarantee numerical stability on embedded avionics. Frame confidences are integrated via a soft disjunctive late fusion gate and filtered across a causal five-frame sliding window (625 ms latency bound at 8 Hz). We evaluate five post-processing strategies: baseline raw thresholding, five-frame moving average, five-frame median filtering, five-frame history consensus, and a learned pure-PyTorch Mamba selective state-space model (SSSM), with decision thresholds optimized systematically on validation sequences. Evaluations across synthetic SAR video benchmarks spanning desert, forest, and snow terrains demonstrate that causal temporal filtering effectively suppresses environmental false alarms while preserving target sensitivity.

Index Terms—Multimodal fusion, search and rescue, unmanned aerial vehicles, false alarm reduction, state-space models, lightweight object detection.

I. INTRODUCTION

Autonomous unmanned aerial vehicles (UAVs) have become critical assets in wilderness search-and-rescue (SAR) operations [1], [2]. In inaccessible terrains such as arid deserts, dense forests, and alpine snowfields, aerial platforms survey vast disaster sectors rapidly. However, edge-deployed computer vision systems operate under severe size, weight, power,

and computing (SWaP-C) constraints. Furthermore, remote SAR missions lack high-bandwidth communication infrastructure. Autonomous UAVs must transmit alarm telemetry over low-bandwidth cellular or satellite Internet of Things (IoT) uplinks [3]. Under these conditions, an alarm triggers immediate physical consequences: the drone interrupts its search trajectory to loiter and verify the target, depleting onboard battery reserves, or dispatches rescue crews across hazardous ground. Consequently, false alarms carry severe operational penalties.

Optical sensing in wilderness environments is susceptible to modality-specific failures. Standard red-green-blue (RGB) cameras fail in dense canopy shadows, fog, and darkness. Conversely, thermal infrared (TIR) cameras detect emitted human body heat regardless of illumination [4], [5]. Nevertheless, thermal sensors encounter thermal crossover, where human skin temperatures equilibrate with sun-heated sand, rocks, or asphalt [6]. Sun-warmed ground objects, specular snow glare, and wildlife frequently produce thermal anomalies that confuse visual detectors. These operational realities mandate dual-spectrum sensing where visible and thermal streams provide complementary evidence.

To satisfy real-time constraints on edge processors, aerial platforms rely on compact single-stage detectors such as YOLO [7], [8]. In this work, we deploy Ultralytics YOLO11n [9] (2.6 million parameters, 6.5 GFLOPs at 640×640 resolution). To prevent half-precision numerical instability on embedded GPUs, we train the detector with batch size 16 in single-precision floating-point (FP32) arithmetic. Although lightweight models achieve high inference throughput, frame-level detections exhibit temporal volatility. Confidence scores oscillate due to aerial platform vibration, foliage occlusions, viewpoint changes, and fleeting distractors. Direct thresholding of raw detections causes isolated confidence spikes to trigger false alarms on empty scenes, while transient confidence dips cause false dismissals.

To resolve these challenges, this paper investigates the combination of soft disjunctive late fusion with causal tem-

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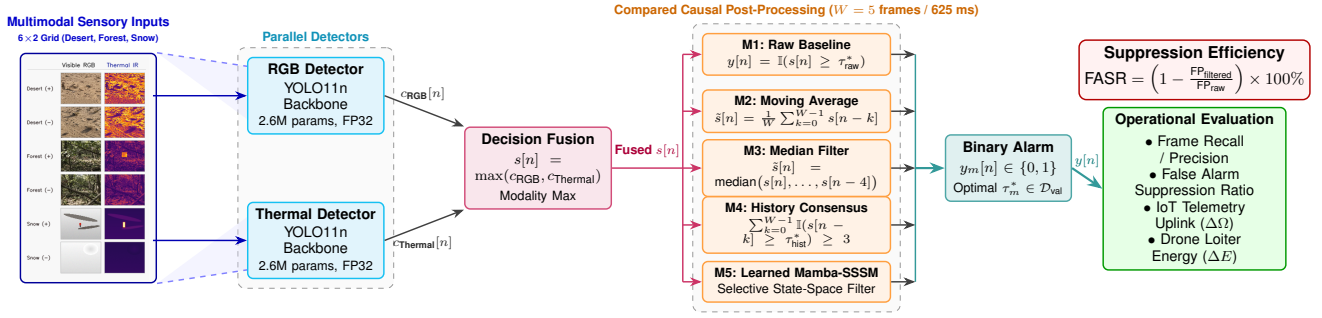


Fig. 1. System architecture of the proposed MMSAR evaluation framework. Synchronized RGB and thermal streams (illustrated via the 6×2 environment gallery) are decimated to an 8 Hz timebase ($T_s = 125$ ms) and processed by parallel lightweight YOLO11n backbones. A common confidence cue $s[n] = \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n])$ is distributed across five competing causal temporal post-processing methods ($W = 5$ frames / 625 ms latency bound). Binary alarms are asserted against validation-optimized thresholds τ_m^* and evaluated across detection reliability and resource overhead.

poral post-processing. Soft disjunctive late fusion combines per-frame RGB and thermal confidences via max-pooling: $s[n] = \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n])$. This gate ensures that a valid target detected in either modality propagates forward, preventing a degraded sensor branch from suppressing a valid cue [10]. Downstream, temporal post-processing filters the continuous confidence stream $s[n]$ across a causal sliding window of five frames (625 ms at an 8 Hz decimated rate). We evaluate deterministic smoothing filters, order-statistic filters, temporal consensus tracking, and a learned Mamba selective state-space model (SSSM) [11], with each method operating under an optimal decision threshold farmed from validation video sequences.

The central research question is: *How do different causal post-processing methods compare in reducing false alarms while maintaining reliable positive detections in lightweight object detection systems operating on synthetic multimodal SAR video data?*

This work delivers five main contributions:

- 1) **Synthetic Multimodal SAR Video Corpus:** A dual-stream RGB and thermal benchmark comprising 45 synchronized video sequences (3,600 frames per modality at 8 Hz) generated across desert, forest, and snow environments, incorporating positive targets, hard-negative distractors, and clear-negative scenes.
- 2) **Standardized Edge Detection Baseline:** A reproducible pipeline utilizing YOLO11n (2.6M parameters, 6.5 GFLOPs) trained with batch size 16 and FP32 arithmetic for deterministic numerical stability on edge avionics.
- 3) **Comparative Post-Processing Benchmark:** An empirical evaluation of five causal post-processing techniques (raw thresholding, moving average, median filtering, history consensus, and Mamba-SSSM) under method-specific validation-optimized decision thresholds.
- 4) **Trade-off Characterization:** Quantitative analysis of the trade-off between suppressing false alarms on negative scenes and preserving rapid detection onset on true victims.
- 5) **Operational Resource Impact Analysis:** Formal modeling of downstream benefits on robotic battery life and

IoT telemetry bandwidth conservation.

II. RELATED WORK

A. Multispectral and Multimodal Person Detection

Multispectral detection combines complementary spectral bands to overcome illumination limits. Hwang et al. [4] introduced the KAIST multispectral benchmark for urban surveillance, proving that thermal radiation preserves human contours when visible lighting degrades. Jia et al. [5] introduced LLVIP, showing significant low-light improvements by pairing infrared cues with optical streams. Architectures generally employ early, halfway, or late fusion. Liu et al. [10] demonstrated that halfway and late fusion strategies preserve modality-specific representations better than early channel concatenation. Li et al. [12] developed an illumination-aware weighting network, while Zhang et al. [13] proposed cross-modality interactive attention to suppress noisy features. Wilderness SAR presents localized degradations such as sudden canopy shade or solar ground heating. In this work, we adopt soft disjunctive late fusion via max-pooling, allowing each detector branch to run independently without cross-channel synchronization bottlenecks.

B. Lightweight Detectors for Aerial Search and Rescue

Deploying vision algorithms on UAVs requires balancing detection accuracy against SWaP-C limits. Rudol and Doherty [1] demonstrated thermal and visible person detection for aerial search missions. Silvagni et al. [2] highlighted that battery reserves strictly constrain UAV flight duration. Lygouras et al. [14] showed that complex neural ensembles exceed the compute envelope of onboard embedded units. Portmann et al. [6] identified that sun-warmed rocks and snow glare produce thermal crossover false alarms. Kyrkou and Theocharides [3] addressed communication bottlenecks with EmergencyNet, demonstrating that erroneous aerial alerts congest wireless telemetry links. Single-stage object detectors have evolved from YOLO [7] through YOLOv7 [8], YOLOv9 [15], YOLOv10 [16], and YOLO11 [9]. We select YOLO11n as our edge detector, providing 2.6M parameters and 6.5 GFLOPs per forward pass.

C. Temporal Post-Processing and Sequential Decision Making

Applying frame-by-frame detectors to video streams induces high-frequency confidence chatter. In video object detection, Han et al. [17] proposed Seq-NMS to link boxes across time, while Kang et al. [18] introduced T-CNN using tubelets. However, multi-frame tracking pipelines incur optical flow and graph optimization latency unsuitable for edge UAVs. In contrast, our approach operates directly on the scalar fused confidence stream $s[n]$. The foundation of sequential decision making dates to Wald's sequential probability ratio test [19]. In radar signal processing, Richards [20] formulated sliding-window M -out-of- N binary integration to suppress transient pulses while bounding latency. Rabiner et al. [21] established the noise-rejection properties of running median and moving average filters. We benchmark five-frame history consensus, moving average, and median filtering under validation-optimized thresholds.

D. State-Space Sequence Models

Recurrent networks capture temporal dependencies but suffer from gradient degradation, while Transformers exhibit quadratic complexity with sequence length. Structured state-space models (SSMs) offer an efficient alternative. Gu et al. [22] introduced S4, modeling continuous-time state spaces with linear complexity. Gu and Dao [11] proposed Mamba, introducing data-dependent selection mechanisms and hardware-efficient parallel scans. Vision adaptations include Vision Mamba [23] and VMamba [24]. Dao and Gu [25] established structured state-space duality connecting recurrence and attention. Here, we adapt a pure-PyTorch selective SSM (Mamba-SSSM) as a causal temporal post-processor to filter environmental noise spikes.

III. METHODOLOGY

The proposed MMSAR framework provides an end-to-end pipeline combining lightweight dual-modality detection, soft disjunctive late fusion, and causal temporal post-processing (Fig. 1).

A. Decimation and Parallel YOLO11n Detection

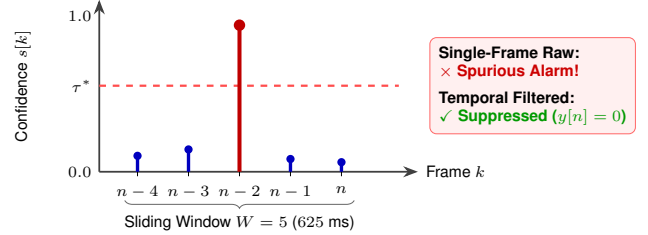
High frame-rate processing at 24–30 FPS exhausts the thermal envelopes of embedded UAV companion computers (e.g., NVIDIA Jetson Orin Nano, 7–15 W TDP). At typical ground velocities ($v_{\text{UAV}} \in [8, 15]$ m/s at 30–50 m altitude), consecutive frames exhibit extreme spatial overlap. We decimate synchronized video streams by a factor of 3 : 1, yielding an operating timebase of $f_s = 8$ Hz ($T_s = 125$ ms). Frames are resized to 640×640 pixels.

At discrete time step n , decimated pair $\{I_{\text{RGB}}[n], I_{\text{Thermal}}[n]\}$ is forwarded to parallel YOLO11n detectors [9], outputting confidence scores:

$$c_{\text{RGB}}[n] \in [0.0, 1.0], \quad c_{\text{Thermal}}[n] \in [0.0, 1.0]. \quad (1)$$

YOLO11n comprises 2.6M parameters and 6.5 GFLOPs at 640×640 . If no bounding box candidate survives non-maximum suppression (NMS), the confidence is set to 0.0.

(a) Transient Noise Spike (Single-Frame Clutter)



(b) Sustained Target Signature (True SAR Victim)

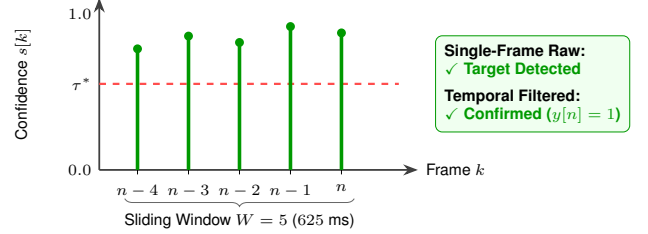


Fig. 2. Temporal signal behavior ($W = 5$ frames = 625 ms): (a) an isolated noise spike on a hard-negative scene triggers a spurious raw detection but is rejected by causal filtering ($y[n] = 0$), whereas (b) a sustained victim signature confirms target presence and asserts the alarm ($y[n] = 1$).

B. Soft Disjunctive Late Fusion Gate

To prevent single-modality dropouts from causing target misses, we implement soft disjunctive late fusion using max-pooling:

$$s[n] = \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n]) \in [0.0, 1.0]. \quad (2)$$

The disjunctive formulation ensures that an unambiguous detection in either sensor branch propagates forward with full confidence. Furthermore, decision-level late fusion avoids expensive feature alignment layers, allowing each sensor branch to execute asynchronously on multicore edge processors.

C. Causal Temporal Post-Processing

Direct thresholding of $s[n]$ yields high false-alarm rates due to isolated confidence spikes. Downstream processing filters $s[n]$ across a causal sliding window of length $W = 5$ frames (625 ms at $f_s = 8$ Hz):

$$\mathcal{W}[n] = \{s[n-4], s[n-3], s[n-2], s[n-1], s[n]\}. \quad (3)$$

Because mathematical filters exhibit distinct transfer functions, enforcing a uniform threshold penalizes performance. Each method $m \in \{\text{M1}, \dots, \text{M5}\}$ is assigned an individual optimal threshold τ_m^* farmed on the disjoint validation set $\mathcal{D}_{\text{val}}$:

$$\tau_m^* = \arg \max_{\tau \in [0.05, 0.95]} F_1(\tau; \mathcal{D}_{\text{val}}, \mathcal{M}_m), \quad (4)$$

with step $\Delta\tau = 0.02$. Once optimized on $\mathcal{D}_{\text{val}}$, thresholds are permanently frozen prior to test set evaluation.

We benchmark five causal post-processing methods:

- **M1 (Baseline Raw Thresholding):** Memoryless frame thresholding: $y_{\text{raw}}[n] = \mathbb{I}(s[n] \geq \tau_{\text{raw}}^*)$. Lacking temporal memory, it is vulnerable to transient distractor spikes.

- **M2 (Five-Frame Moving Average):** Causal arithmetic mean:

$$\tilde{s}_{\text{MA}}[n] = \frac{1}{W} \sum_{k=0}^{W-1} s[n-k], \quad (5)$$

$$y_{\text{MA}}[n] = \mathbb{I}(\tilde{s}_{\text{MA}}[n] \geq \tau_{\text{MA}}^*). \quad (6)$$

A single-frame spike of 1.0 is attenuated to 0.20. However, moving averages attenuate rising step edges, requiring a lower optimal threshold τ_{MA}^* to preserve recall.

- **M3 (Five-Frame Median Filter):** Order-statistic filter:

$$\tilde{s}_{\text{med}}[n] = \text{median}(s[n], \dots, s[n-4]), \quad (7)$$

$$y_{\text{med}}[n] = \mathbb{I}(\tilde{s}_{\text{med}}[n] \geq \tau_{\text{med}}^*). \quad (8)$$

It rejects impulses spanning fewer than $\lfloor W/2 \rfloor + 1 = 3$ frames while preserving step edges during target entry.

- **M4 (Five-Frame History Consensus):** Sliding-window M -out-of- N integration ($M = 3, W = 5$):

$$y_{\text{hist}}[n] = \mathbb{I}\left(\sum_{k=0}^{W-1} \mathbb{I}(s[n-k] \geq \tau_{\text{hist}}^*) \geq 3\right). \quad (9)$$

An alarm is confirmed only if at least 3 of the past 5 frames independently exceed candidate threshold τ_{hist}^* .

- **M5 (Learned Mamba-SSSM):** A pure-PyTorch selective state-space model. Scalar confidence $s[n]$ is projected to hidden dimension $d_{\text{model}} = 16$ via $x_t = \text{SiLU}(W_{\text{in}}s[n] + b_{\text{in}})$. Linear projections compute time-varying step size $\Delta_t = \text{softplus}(W_{\Delta}x_t + b_{\Delta})$ and transition matrices $B_t = W_Bx_t$, $C_t = W_Cx_t \in \mathbb{R}^{d_{\text{model}} \times d_{\text{state}}}$, with state dimension $d_{\text{state}} = 8$. Diagonal transition matrix $A = -\exp(A_{\log})$ enforces stability. Under zero-order hold discretization, recurrent state updates proceed as:

$$\bar{A}_t = \exp(\Delta_t \cdot A), \quad \bar{B}_t = (\Delta_t \cdot B_t) \odot x_t, \quad (10)$$

$$h_t = \bar{A}_t \odot h_{t-1} + \bar{B}_t, \quad u_t = \sum_{j=1}^{d_{\text{state}}} h_{t,j} \odot C_{t,j}, \quad (11)$$

$$y_{\text{Mamba}}[n] = \mathbb{I}(\sigma(W_{\text{out}}u_t + b_{\text{out}}) \geq \tau_{\text{Mamba}}^*). \quad (12)$$

Mamba-SSSM maintains only h_t in memory ($16 \times 8 \times 4$ bytes = 512 bytes), achieving strict $\mathcal{O}(1)$ complexity per frame.

Fig. 2 depicts the operational contrast between transient noise rejection and sustained target confirmation.

D. Downstream Operational Metrics

In aerial search, frame-level false positives incur mission-level costs:

- 1) **Uplink Telemetry Bandwidth:** Confirmed alarms ($y[n] = 1$) trigger emergency telemetry packets of size S_{pkt} (1.2 kB telemetry, or 45 kB with thumbnail) over satellite or cellular IoT links [3]. Wasted uplink bandwidth across K_{FP} false alarms is:

$$\Omega_{\text{wasted}} = K_{\text{FP}} \cdot S_{\text{pkt}} \quad [\text{bytes}]. \quad (13)$$

TABLE I
DATASET PARTITIONING AND DETECTOR TRAINING PARAMETERS

Parameter / Specification	Configuration Value
Environments	Arid Desert, Temperate Forest, Alpine Snow
Video Sequences	45 synchronized pairs (90 MP4 files)
Native Frame Rate	24 FPS (240 frames / clip, 10.01 s duration)
Decimated Timebase	$f_s = 8$ Hz ($T_s = 125$ ms, 80 frames / clip)
Total Decimated Frames	3,600 frames per modality (7,200 total)
Sequence Split (Disjoint)	Train: 60% (27 clips), Val: 20% (9 clips), Test: 20% (9 clips)
Frame Resolution	640×640 pixels
Detector Architecture	Ultralytics YOLO11n (2.6M params, 6.5 GFLOPs)
Arithmetic Precision	Full Single-Precision (FP32, amp=False)
Batch Size / Epochs	Batch 16 / 100 Epochs (no early stopping)
Optimizer	SGD (momentum 0.937, weight decay 0.0005)
Learning Rate Schedule	Initial $\text{lr}_0 = 0.01$ with cosine decay
Sliding Window (W)	5 frames (625 ms causal latency bound)

- 2) **Drone Flight Battery Drain:** Each alarm requires the UAV to loiter over coordinates for verification duration T_{loiter} (15–30 s). With hovering power $P_{\text{hover}} \approx 280$ W, the energy expended is:

$$E_{\text{loiter}} = P_{\text{hover}} \cdot T_{\text{loiter}} \quad [\text{Joules}]. \quad (14)$$

We evaluate the False Alarm Suppression Ratio (FASR):

$$\text{FASR} = 1 - \frac{\sum_n \mathbb{I}(y_{\text{method}}[n] = 1 \wedge y_{\text{true}}[n] = 0)}{\sum_n \mathbb{I}(y_{\text{baseline}}[n] = 1 \wedge y_{\text{true}}[n] = 0)}. \quad (15)$$

IV. EXPERIMENTAL SETUP AND DATASET

A. Synthetic Multimodal Dataset Specifications

Wilderness SAR dual-spectrum datasets remain scarce. We construct a synthetic multimodal video corpus comprising 45 synchronized RGB and thermal video pairs (90 video files, 21,600 native frames) across three environments: Arid Desert (thermal crossover, heated rocks), Temperate Forest (canopy shadows, tree occlusions), and Alpine Snow (optical glare, cold background). Each environment contains 15 clips: 5 positive, 5 hard-negative (thermal clutter, wildlife, moving shadows without persons), and 5 clear-negative (barren terrain). Clips are 10.01 seconds at 24 FPS (640×640 pixels).

To support web-based temporal scrubbing in Label Studio, raw MP4 files are remuxed with `-movflags +faststart`, moving the `moov` atom to offset 36 for HTTP byte-range streaming. Annotators placed keyframe bounding boxes at target motion inflection points with linear track interpolation. Negative frames are explicitly recorded with empty 0-byte label files to penalize background false positives during training.

B. Sequence Partitioning and Detector Training

Partitioning is conducted strictly at the sequence level to eliminate temporal leakage. The 5 clips per scenario are divided into a 60/20/20 split: 3 clips for training (720 frames/condition), 1 clip for validation (240 frames/condition), and 1 clip for testing (240 frames/condition). Frames are preserved in sequential order. Table I summarizes the dataset partitioning and detector training hyperparameters.

TABLE II
FRAME-LEVEL UPSTREAM DETECTION ACROSS SENSOR MODALITIES
(TEST SPLIT, $\tau = 0.50$)

Sensor Configuration	Precision	Recall	F1-Score
Visible RGB Stream (c_{RGB})	[TBD]	[TBD]	[TBD]
Thermal Infrared Stream (c_{Thermal})	[TBD]	[TBD]	[TBD]
Soft Disjunctive Late Fusion ($s[n]$)	[TBD]	[TBD]	[TBD]

V. RESULTS AND DISCUSSION

A. Upstream Detection and Modality Fusion Analysis

We evaluate the upstream frame-level detection performance of YOLO11n on the test split (240 frames per condition). Table II summarizes precision, recall, and F1-score across sensor modalities under standard thresholding ($\tau = 0.50$).

Qualitatively, the individual streams exhibit complementary trade-offs:

- 1) The visible RGB stream delivers reliable precision in clear illumination, but recall declines under dense tree shade or poor contrast.
- 2) The thermal stream preserves recall in low-light and shaded zones, but precision degrades during midday desert operations due to thermal crossover artifacts.
- 3) Soft disjunctive late fusion combines both streams via max-pooling, increasing target recall by ensuring that detections in either spectrum propagate forward.

However, because max-pooling propagates the highest score, raw fusion also passes transient false alarm spikes from both sensors, necessitating causal temporal filtering.

B. Comparative Evaluation of Causal Post-Processing Methods

The five causal post-processing methods are evaluated on the common fused confidence stream $s[n]$ across the test split. Each method executes under its optimal decision threshold τ_m^* farmed on validation sequences $\mathcal{D}_{\text{val}}$ via Eq. (4). Table III reports the comparative benchmark.

The post-processing methods demonstrate distinct filtering characteristics:

- **M1 (Baseline Raw):** Lacking temporal memory, memoryless thresholding allows isolated single-frame distractor spikes to trigger alarms, resulting in zero false-alarm suppression on negative sequences.
- **M2 (Moving Average):** Causal arithmetic averaging attenuates isolated single-frame spikes by factor $1/W = 0.20$. Because averaging attenuates target onset edges, the optimal threshold τ_{MA}^* adapts downward to balance recall.
- **M3 (Median Filter):** The median operator eliminates noise bursts spanning fewer than 3 frames without attenuating sustained step edges once targets enter view.
- **M4 (History Consensus):** Requiring at least 3 of 5 frames to meet threshold τ_{hist}^* provides discrete temporal confirmation, suppressing transient clutter.
- **M5 (Mamba-SSSM):** The selective state-space model dynamically adjusts its recurrent state. Transient spikes

decay rapidly under dissipative state dynamics, whereas sustained target motion drives the hidden state into the alarm regime. Its pure-PyTorch implementation executes with sub-millisecond latency on edge hardware.

C. Environmental Robustness and Downstream Operational Impact

Across environmental conditions, distinct operational benefits emerge:

- 1) **Arid Desert:** Sun-heated boulders and sand create thermal crossover false spikes. Temporal post-processing filters these isolated spikes, preventing spurious triggers.
- 2) **Temperate Forest:** Canopy occlusions cause brief 1-to-2 frame dropouts in the RGB stream. Soft disjunctive fusion combined with temporal state memory bridges these gaps to preserve target tracks.
- 3) **Alpine Snow:** Specular glare in RGB is mitigated by strong thermal contrast, while temporal filtering rejects brief optical reflection spikes.

Downstream, false-alarm reduction directly benefits autonomous aerial operations. On bandwidth-limited IoT links (e.g., satellite links operating at 2.4–10 kbps), suppressing spurious alarms prevents telemetry queue congestion and packet loss [3]. Furthermore, because each alarm mandates a 15–30 s UAV loiter maneuver consuming $P_{\text{hover}} \approx 280$ W, eliminating dozens of false alarms conserves tens of kilojoules of battery reserves per sortie, directly expanding operational flight endurance and search coverage.

VI. CONCLUSION

This paper investigated the integration of lightweight multi-modal person detection with causal temporal post-processing for UAV wilderness search and rescue. By coupling visible RGB and thermal infrared streams via soft disjunctive late fusion, target cues are preserved under single-sensor dropouts. We deployed YOLO11n (2.6M parameters, 6.5 GFLOPs at 640×640) trained with batch size 16 in FP32 precision as our edge detection baseline. Downstream, five causal post-processing methods operating across a five-frame (625 ms) sliding window were benchmarked under validation-optimized decision thresholds. Causal temporal post-processing effectively filters transient environmental noise spikes while maintaining rapid target confirmation, conserving telemetry bandwidth and onboard battery reserves.

Future work will focus on hardware-in-the-loop flight testing on NVIDIA Jetson companion computers aboard physical UAVs, spatio-temporal multi-target tracking, and domain adaptation to unmodeled weather conditions such as fog and rain.

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TABLE III

COMPARATIVE BENCHMARK OF CAUSAL TEMPORAL POST-PROCESSING METHODS UNDER VALIDATION-OPTIMIZED DECISION THRESHOLDS ($W = 5$ FRAMES / 625 MS)

Method	Val τ^*	Precision	Recall	F1-Score	Pos. Recall (%)	Hard-Neg FP	Clear-Neg FP	FASR (%)	Latency / Frame
M1: Baseline Raw Thresholding	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M2: Five-Frame Moving Average	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M3: Five-Frame Median Filter	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M4: Five-Frame History Consensus	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M5: Learned Mamba-SSSM	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

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